



List Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Tuples and Sets.

Section 1: List Basics and Creation

★ Create and Display a List

★ Access Elements using Indexing

★ Negative Indexing Practice

★ List Slicing Basics

Reverse a List using Slicing

★ Find Length of a List

Nested List Creation and Access

List of Mixed Data Types

Section 2: Adding and Removing Elements

★ Append an Element to a List

★ Insert Element at a Specific Position

Extend a List with Another List

★ Remove an Element by Value

★ Remove an Element by Index using pop()

Delete Element using del Statement

★ Clear All Elements from a List

Remove Duplicates from a List

Section 3: Searching and Counting

★ Search an Element using in Operator

★ Find Index of an Element

Count Occurrences of an Element

★ Find Maximum and Minimum in a List

Find Second Largest Element

Find Second Smallest Element

Count Even and Odd Numbers in a List

Count Positive, Negative, and Zero in a List

Section 4: Sorting and Ordering

★ Sort a List in Ascending Order

★ Sort a List in Descending Order



Sort List of Strings Alphabetically

Sort List without using sort() (Bubble Sort)

★ **Sort List using Custom Key Function**

Sort List of Tuples by Second Element

Check if a List is Sorted

Section 5: List Manipulation

★ **Concatenate Two Lists**

★ **Repeat a List N Times**

Merge Two Lists Alternately

Swap Two Elements in a List

★ **Rotate a List Left by K Positions**

Rotate a List Right by K Positions

Flatten a Nested List

Split a List into Chunks of Size N

Section 6: List Comprehension

★ **Squares of Numbers using List Comprehension**

★ **Filter Even Numbers using List Comprehension**

Generate List of Prime Numbers using Comprehension

★ **Convert List of Strings to Uppercase**

Flatten Nested List using Comprehension

Create Pairs using Nested Comprehension

Conditional Expression inside Comprehension

Section 7: List Aggregation and Statistics

★ **Sum of All Elements in a List**

★ **Average of Elements in a List**

Product of All Elements in a List

Frequency of Each Element (without dict)

★ **Find Common Elements between Two Lists**

Find Elements in One List but Not in Another

Find Union of Two Lists

Section 8: List-Based Logical Problems

★ **Check if a List is a Palindrome**

★ **Remove Duplicates while Preserving Order**

Find Missing Number in a Sequence

★ **Move All Zeros to End of List**

Find Pairs with a Given Sum



★ Find the Intersection Count of Two Lists

Cyclic Rotation Checker

Find All Sublists of a Given List

Section 9: Real World List Applications

★ Student Marks List Analyzer

★ Shopping Cart Total Calculator

Attendance Tracker using List

★ Employee Salary List Processor

Inventory Stock Manager

To-Do List Manager using List Operations